**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE**

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TEST CASES & VALIDATION REPORT

# Solana Local Explorer

92 test cases across six levels — 38 executed automatically, 54 documented as reproducible procedures — with a full requirements traceability matrix.

AUTHORS

**Garvit Dadheech · Shubh Kesharwani**

SUPERVISOR

**Dharma Teja**

| 92 | 38 | 38 | 0 | 2 |
|---|---|---|---|---|
| TEST CASES | AUTOMATED | PASSED | FAILED | OPEN DEFECTS |

This report is also reproduced as Chapter 6 and Appendix D of the Final Project Report. · Automated run: 28 August 2026

# 1. Test Strategy

Testing was organised around a simple principle: **automate what can be verified without infrastructure, and script precisely what cannot**. The two layers where a silent bug would corrupt data — gRPC payload parsing and 64-bit serialisation — are pure functions, so they are covered by an automated suite that runs in under a second with no validator and no database. Everything above them depends on a live four-container stack, so those cases are written as reproducible procedures with explicit commands, preconditions and expected results.

> **HOW TO READ THE RESULT COLUMNS**
>
> Cases marked AUTOMATED were executed by the authors with `pnpm test` on **28 August 2026**; the verbatim console output is reproduced in Section 11 and can be regenerated by anyone in seconds. Cases marked MANUAL are scripted procedures exercised against the running stack during development; they are documented here in full so that an examiner can reproduce each one independently rather than take the result on trust. Only the automated results constitute machine-checkable evidence.

| LEVEL | METHOD | TOOLING | CASES | WHAT IT PROTECTS |
|---|---|---|---|---|
| Unit — parsing | AUTOMATED | `Vitest` | 29 | Correct decoding of every field of the gRPC transaction payload, including malformed and missing input. |
| Unit — serialisation | AUTOMATED | `Vitest` | 9 | Lossless conversion of `BigInt` and `Date` to JSON, with ordinary numbers untouched. |
| Integration | MANUAL | `docker · psql` | 12 | Container health, start-up ordering, migration, stream subscription, end-to-end persistence. |
| API | MANUAL | `curl · jq` | 18 | Endpoint contracts, pagination, filtering, search, error codes and JSON encoding. |
| Interface | MANUAL | `browser` | 16 | Navigation, live refresh, detail rendering, filter behaviour, empty and error states. |
| Non-functional | MANUAL | `mixed` | 8 | Latency, restart behaviour, idempotence, portability, reset and configurability. |
| **Total** | | | **92** | **38 automated · 54 documented manual procedures** |

**Table 1** — Test levels, tooling and coverage.

## 2. Test Environment

| | |
|---|---|
| HOST | Apple Silicon macOS workstation; validator running under `linux/amd64` emulation |
| CONTAINER RUNTIME | Docker Desktop 24.x with Compose V2 |
| AUTOMATED RUNNER | Vitest 4.1.11 on Node.js 22, pnpm 10.27 |
| SERVICES UNDER TEST | `solana-validator` · `explorer-postgres` · `solana-indexer` · `solana-explorer` |
| DATABASE ACCESS | `postgresql://postgres:password@localhost:5433/explorer` |
| EXPLORER UNDER TEST | `http://localhost:3000` |
| TEST DATA | Validator vote transactions produced continuously by the test validator, plus deliberate transfers created with the Solana CLI inside the validator container |
| DATE OF AUTOMATED RUN | 28 August 2026 |

## 3. Unit Test Results

| 38 | 38 | 0 | 2 | <1 s |
|---|---|---|---|---|
| AUTOMATED CASES | PASSED | FAILED | TEST FILES | SUITE RUNTIME |

### 3.1 Explorer — `serializeBigInts()`

Nine cases cover the function's entire behavioural surface: pass-through, conversion, recursion and the two edge values (negative and zero) that a naive truthiness check would mishandle.

| ID | CASE | EXPECTED RESULT | STATUS |
|---|---|---|---|
| UT-SER-01 | Returns null / undefined unchanged | Null and undefined pass through without conversion or error | PASS |
| UT-SER-02 | Converts a top-level bigint to a string | `123n → "123"` | PASS |
| UT-SER-03 | Converts a Date to an ISO string | `"2026-01-01T00:00:00.000Z"` | PASS |
| UT-SER-04 | Leaves primitives unchanged | Number, string and boolean are returned identically and keep their types | PASS |
| UT-SER-05 | Recursively converts bigints in a nested object | Every bigint at any depth becomes a string; other keys untouched | PASS |
| UT-SER-06 | Recursively converts bigints inside an array | Each element is processed independently and order is preserved | PASS |
| UT-SER-07 | Realistic API payload (transaction list + pagination) | `slot` and `fee` become strings, `blockTime` becomes ISO, and `pagination.page` remains a number | PASS |
| UT-SER-08 | Negative bigint (balanceChange can be negative) | `-10000n → "-10000"` | PASS |
| UT-SER-09 | Zero as a bigint | `0n → "0"`, not dropped as falsy | PASS |

**Table 2** — Explorer serialisation unit tests (`explorer/src/lib/serialization.test.ts`).

### 3.2 Indexer — gRPC payload parsing

Twenty-nine cases cover each extractor with a valid buffer, a typed array, an already-decoded string, a missing field and — where relevant — a value of an unexpected type, plus one end-to-end case over a complete realistic payload.

| ID | UNIT | CASE | EXPECTED RESULT | STATUS |
|---|---|---|---|---|
| UT-PRS-01 | `parseSignature` | Buffer signature | Base-58 encoding of the 64-byte buffer | PASS |
| UT-PRS-02 | `parseSignature` | Uint8Array signature | Identical result to the Buffer case | PASS |
| UT-PRS-03 | `parseSignature` | Already-decoded string | Returned unchanged, not re-encoded | PASS |
| UT-PRS-04 | `parseSignature` | Missing (null / undefined) | `"unknown"`, no exception | PASS |
| UT-PRS-05 | `parseSignature` | Unexpected type (number) | `"unknown"`, no exception | PASS |
| UT-PRS-06 | `parseAccounts` | Buffer account keys | Array of base-58 strings | PASS |

| ID | UNIT | CASE | EXPECTED RESULT | STATUS |
|---|---|---|---|---|
| UT-PRS-07 | parseAccounts | String account keys | Returned unchanged | PASS |
| UT-PRS-08 | parseAccounts | Missing accountKeys | Empty array, no exception | PASS |
| UT-PRS-09 | parseAccounts | Order preservation | Input order preserved — the index is the balance-row key | PASS |
| UT-PRS-10 | parseInstructions | Full instruction | Program index, account list, base-58 data and ordinal all captured | PASS |
| UT-PRS-11 | parseInstructions | Missing accounts field | Defaults to an empty array | PASS |
| UT-PRS-12 | parseInstructions | Instruction with no data | `data` is null rather than undefined or "" | PASS |
| UT-PRS-13 | parseInstructions | Multiple instructions | Ordinals are 0, 1, 2 matching array position | PASS |
| UT-PRS-14 | parseInstructions | Missing instructions | Empty array, no exception | PASS |
| UT-PRS-15 | parseComputeUnitsUsed | Value present | `2100 → 2100n` | PASS |
| UT-PRS-16 | parseComputeUnitsUsed | Value missing | `undefined` | PASS |
| UT-PRS-17 | parseComputeUnitsUsed | Value is 0 (falsy but valid) | `undefined` — documents defect D-06 (Section 8) rather than asserting desired behaviour | PASS |
| UT-PRS-18 | parseFee | Fee present | `5000 → 5000n` | PASS |
| UT-PRS-19 | parseFee | Fee missing | `undefined` | PASS |
| UT-PRS-20 | parseSuccess | `meta.err === null` | `true` | PASS |
| UT-PRS-21 | parseSuccess | `meta.err` absent | `true` — absence is not failure | PASS |
| UT-PRS-22 | parseSuccess | Populated error object | `false` | PASS |
| UT-PRS-23 | parseError | No error | `null` | PASS |
| UT-PRS-24 | parseError | Error object present | Serialised with `JSON.stringify` for storage as text | PASS |
| UT-PRS-25 | parseBalances | Both arrays present | Pre- and post-balance arrays extracted intact | PASS |
| UT-PRS-26 | parseBalances | Arrays missing | Two empty arrays, so the balance loop is skipped safely | PASS |
| UT-PRS-27 | parseLogs | Log messages present | Array returned intact and in order | PASS |
| UT-PRS-28 | parseLogs | Log messages missing | Empty array | PASS |
| UT-PRS-29 | end-to-end | Complete realistic payload | All ten fields parse consistently from one vote-transaction-shaped message | PASS |

**Table 3** — Indexer parsing unit tests (`indexer/src/grpc/parse-transaction.test.ts`).

**A NOTE ON THE TEST HELPER**

The parsing logic lives inline in the `stream.on("data", …)` handler in `indexer.ts`. To make it testable without restructuring the production hot path mid-project, the suite exercises a documented mirror of that logic in `parse-transaction.test-helper.ts`, which carries a header warning that it must be updated in step with the handler. This is an acknowledged trade-off: it buys coverage now at the cost of a manual synchronisation obligation. Extracting the handler into exported functions and pointing the tests at the real implementation is listed as future work in the project report, Section 9.2.

## 4. Integration Test Cases

| ID | OBJECTIVE | PROCEDURE | EXPECTED RESULT |
|---|---|---|---|
| IT-01 | Clean build from source | `docker compose up --build -d` on a machine with no cached layers | All four images build; the validator stage compiles the Geyser plugin without error; no manual intervention required. |
| IT-02 | All services reach running state | `docker compose ps` | Four rows `Up`; `solana-validator` and `explorer-postgres` additionally show `(healthy)`. |
| IT-03 | Validator RPC responds | `curl -X POST localhost:8899 -d '{"jsonrpc":"2.0","id":1,"method":"getSlot"}'` | HTTP 200 with a JSON body containing a numeric `result` that increases on repeat calls. |
| IT-04 | Health gate is honoured | Inspect start-up order in `docker compose logs` | The indexer's first log line appears only after both health checks pass; no connection-refused errors precede it. |
| IT-05 | Migrations apply on first start | `docker compose logs indexer \| head` | "Running database migrations…" followed by "Migrations completed"; all three migrations applied. |
| IT-06 | Migration retry under a slow database | Start the stack on a cold machine where PostgreSQL initialises slowly | The indexer logs a retry and succeeds on a later attempt rather than exiting; the container does not enter a crash loop. |
| IT-07 | gRPC subscription established | `docker compose logs indexer` | "Connecting to Yellowstone gRPC: http://validator:10000" followed by "Subscribed to slots and transactions". |
| IT-08 | Slot updates received | Observe indexer logs for 30 seconds | Continuous `Slot <n> status: <s>` lines with a monotonically increasing slot number. |
| IT-09 | End-to-end transaction persistence | Airdrop and transfer inside the validator container, then query the database | A row with the new signature appears in `Transaction`; the indexer logs `Indexed transaction <prefix>…`. |

| ID | OBJECTIVE | PROCEDURE | EXPECTED RESULT |
|---|---|---|---|
| IT–10 | Balance rows written per account index | `SELECT * FROM "AccountBalance" WHERE "transactionId" = …` | One row per account key, with `accountIndex` matching wire order and `balanceChange = postBalance − preBalance`. |
| IT–11 | Account registry and links populated | Query `Account` and `AccountTransaction` | Every address in the transaction has an `Account` row and exactly one join row per transaction. |
| IT–12 | Failed transaction captured separately | Submit a transaction that fails (for example, transfer more than the balance) | A row appears in `FailedTransaction` with the serialised error and the program logs; `Transaction` is unaffected; the dashboard's failed count increments. |

**Table 4** — Integration test cases.

## 5. API Test Cases

| ID | REQUEST | EXPECTED RESPONSE |
|---|---|---|
| API-01 | `GET /api/stats` | 200; body contains `totalTransactions`, `totalFailed`, `successRate`, `latestSlot` and a `recentTransactions` array of at most 10. |
| API-02 | `GET /api/stats` | `successRate` equals `total ÷ (total + failed) × 100` rounded to one decimal; equals 0 when both counts are 0 (no division by zero). |
| API-03 | `GET /api/stats` | `latestSlot` is a decimal string, not a number, and equals the maximum indexed slot. |
| API-04 | `GET /api/transactions` | 200; 20 items by default; `pagination` reports `page 1`, `limit 20` and a correct `totalPages`. |
| API-05 | `GET /api/transactions?limit=5&page=2` | Exactly 5 items, disjoint from page 1, ordered by descending slot. |
| API-06 | `GET /api/transactions?limit=5000` | Limit is clamped to 100; the response never exceeds 100 items. |
| API-07 | `GET /api/transactions?page=0&limit=0` | Page floors to 1 and limit floors to 1; no error and no empty page. |
| API-08 | `GET /api/transactions?page=abc` | Unparseable values fall back to defaults rather than producing `NaN` in the query. |
| API-09 | `GET /api/transactions/{valid}` | 200; full record including `accounts`, `instructions` and `memos`; `slot` and `fee` are strings. |
| API-10 | `GET /api/transactions/{failed}` | 200; the failed record is returned with `success: false`, the error string and the log array. |
| API-11 | `GET /api/transactions/{unknown}` | 404 with `{"error":"Transaction not found"}`. |
| API-12 | `GET /api/accounts/{known}` | 200; account record, `latestBalance`, `totalBalanceChange`, a balance array and a pagination envelope. |
| API-13 | `GET /api/accounts/{unknown}` | 404 with `{"error":"Account not found"}`. |
| API-14 | `…?field=postBalance&amount=1&unit=sol` | Only rows whose post-balance is exactly 1 000 000 000 lamports; `filter` echoes the applied criteria; count and pagination reflect the filtered set. |
| API-15 | `…?minAmount=0.1&maxAmount=10` | Inclusive range applied to the selected field; an exact `amount`, if also supplied, takes precedence over the range. |
| API-16 | `…?field=DROP TABLE` | Unrecognised field silently falls back to `postBalance`; the value never reaches the query builder. |
| API-17 | `GET /api/search?q=abcdefgh` | 200; up to three signature matches and three address matches by prefix, each tagged with its type. |
| API-18 | `GET /api/search?q=abc` | 200 with an empty `results` array — the eight-character minimum is enforced server-side, not only in the UI. |

**Table 5** — API test cases.

## 6. User Interface Test Cases

| ID | SCENARIO | EXPECTED BEHAVIOUR |
|---|---|---|
| UI-01 | Dashboard first paint | Skeletons matching the final layout are shown, then replaced by the four stat cards, the chart and the recent-transactions table. |
| UI-02 | Dashboard live refresh | Values update roughly every five seconds with no full-page reload and no skeleton flash on subsequent refreshes. |
| UI-03 | Backend unavailable | The dashboard renders an explicit “Failed to load stats data” panel rather than an empty page or a stack trace. |
| UI-04 | Activity chart ranges | Switching 5 m / 15 m / 1 h / All re-buckets client-side and immediately; successes and failures are stacked and separately coloured. |
| UI-05 | Chart with no data | An explanatory empty state is shown instead of an axis with no series. |
| UI-06 | Transaction list pagination | Next / previous update both the table and the URL; a reload restores the same page. |
| UI-07 | Signature truncation | Truncated signatures reveal the full value on hover and can be copied with the inline control. |
| UI-08 | Navigate to transaction detail | Selecting a signature routes to `/transactions/{sig}` and renders status, slot, fee, compute units, accounts and instructions. |
| UI-09 | Failed transaction detail | The error is displayed prominently and the program logs are rendered in a scrollable monospace panel. |
| UI-10 | Unknown signature | A “Transaction not found” state with a route back to the transaction list; no unhandled error. |
| UI-11 | Account link from a transaction | Each account in the account list links to `/accounts/{address}` and navigates correctly. |
| UI-12 | Account detail rendering | Address, first seen, last seen, current balance in SOL to nine decimals, and the paginated balance history are all shown. |
| UI-13 | Balance change formatting | Positive changes are prefixed with + and coloured as gains; negative changes are coloured as losses; zero is neutral. |
| UI-14 | Balance filter round trip | Applying a filter updates the URL; reloading preserves it; “Clear” returns to the unfiltered view. |
| UI-15 | ⌘K search | The dialog opens on ⌘K / Ctrl-K, debounces input, lists matches grouped by type, and routes to the chosen record. |
| UI-16 | Theme switching | Light, dark and system themes apply immediately across every page and persist across reloads. |

**Table 6** — User interface test cases.

## 7. Non-Functional Test Cases

| ID | ATTRIBUTE | PROCEDURE | ACCEPTANCE CRITERION |
|---|---|---|---|
| NF-01 | Usability (NFR-1) | A user with only Docker follows the README Quick Start | A working explorer is reached with one command and no undocumented prerequisite. |
| NF-02 | Latency (NFR-2) | Send a transfer and watch the dashboard | The transaction appears within one five-second refresh cycle. |
| NF-03 | Reliability (NFR-3) | `docker restart solana-indexer` | The container restarts, re-runs migrations as a no-op, re-subscribes and resumes indexing without operator action. |
| NF-04 | Idempotence (NFR-4) | Restart the indexer and let the stream redeliver | No duplicate signatures: `SELECT signature, COUNT(*) … HAVING COUNT(*) > 1` returns no rows. |
| NF-05 | Precision (NFR-4) | Compare a large lamport balance in the API response with the database value | Exact match; the value is a string in JSON and no digits are lost. |
| NF-06 | Portability (NFR-6) | Build and run on Apple Silicon under `linux/amd64` emulation | The stack builds and runs unmodified; only build time is affected. |
| NF-07 | Reset (FR-22) | `docker compose down -v` then `up -d` | Both named volumes are removed; the next start yields a fresh ledger and an empty, freshly migrated database. |
| NF-08 | Configurability (NFR-8) | Start with `YELLOWSTONE_ADDR` pointing at an external endpoint | The indexer connects to the external endpoint — confirmed in the log line — and indexes that chain instead, with no code change. |

**Table 7** — Non-functional test cases.

## 8. Defects Found and Resolved

The following defects were found during development and testing. Eight were fixed; two are documented open items with known workarounds, recorded here rather than omitted.

| ID | DEFECT | SEVERITY | ROOT CAUSE AND RESOLUTION | STATE |
|---|---|---|---|---|
| D-01 | `JSON.stringify` throws on Prisma `BigInt` fields | Critical | PostgreSQL `BIGINT` maps to JS `BigInt`, which JSON cannot encode; every endpoint returning a slot or a fee failed. Fixed by introducing `serializeBigInts()` and applying it at every response boundary; pinned by nine unit tests. | **Fixed** |
| D-02 | gRPC client rejects a `host:port` endpoint | High | The Yellowstone client parses a full URL. A bare `validator:10000` produced a connection failure at start-up. Fixed by normalising the value with an `http://` prefix when no scheme is present. | **Fixed** |
| D-03 | Stream terminates intermittently under load | High | The default gRPC maximum receive length is smaller than a busy slot's payload. Fixed by setting `grpc.max_receive_message_length` to 64 MB on the client. | **Fixed** |
| D-04 | Indexer exits on a cold start because the database is not ready | High | Compose health checks confirm that PostgreSQL accepts connections but not that first-time initialisation is complete, so `migrate deploy` could still fail. Fixed with a bounded retry loop (10 attempts, 3 s apart) before exiting non-zero. | **Fixed** |
| D-05 | Explorer image build fails at dependency install | High | The `postinstall` hook runs `prisma generate`, which needs the schema, but the Dockerfile copied only `package.json` and the lockfile into the dependency stage. Fixed by copying `prisma/` before install (commit `a8b88ad`). | **Fixed** |
| D-06 | A transaction consuming exactly 0 compute units is stored as “unknown” | Low | The extractor uses a truthiness check, and `0` is falsy in JavaScript, so a genuine zero is indistinguishable from a missing field. The behaviour is deliberately pinned by UT-PRS-17 with an explanatory comment so it cannot regress silently. *Open*: the fix is an explicit `!== undefined` check; deferred to avoid a late change to the ingestion hot path. | **Open** |
| D-07 | Redundant sequential queries on the account endpoint | Medium | Balance rows, count, latest balance and aggregate were fetched one after another, so latency was the sum of four round trips. Fixed by consolidating them into a single `Promise.all` (commit `9e7f322`). | **Fixed** |
| D-08 | Prisma client constructor fails under `tsx` | Medium | A Prisma/`tsx` interaction caused the no-argument constructor to fail in the indexer's development runner. Fixed with a guarded fallback that retries with an explicit empty options object, documented in `db/client.ts`. | **Fixed** |
| D-09 | Account page reports “Account not found” for previously seen addresses | Low | Balance tracking was introduced in the third migration, so addresses last seen before that point have no `AccountBalance` rows. *Open by design*: documented in the README and the user manual; resolved for any address involved in a subsequent transaction, or by a full reset. | **Open** |
| D-10 | Non-ASCII characters in indexer log output | Cosmetic | Emoji in log lines rendered inconsistently across terminals and in captured CI logs. Fixed by replacing them with plain text (commit `739fa24`). | **Fixed** |

**Table 8** — Defect log.

## 9. Requirements Traceability Matrix

Every functional and non-functional requirement maps to at least one test case.

| REQUIREMENT | VERIFIED BY | REQUIREMENT | VERIFIED BY |
|---|---|---|---|
| FR-1 | IT-07, IT-08, NF-08 | FR-12 | IT-05, IT-06, NF-03 |
| FR-2 | IT-07, NF-08 (defect D-02) | FR-13 | API-01, API-02, API-03, UI-01 |
| FR-3 | NF-08 | FR-14 | API-04 – API-08, UI-06 |
| FR-4 | UT-PRS-01 – 09, UT-PRS-29 | FR-15 | API-09, API-10, API-11, UI-08 – UI-10 |
| FR-5 | UT-PRS-10 – 14 | FR-16 | API-12, API-13, UI-12 |
| FR-6 | UT-PRS-15 – 24, IT-12 | FR-17 | API-14, API-15, API-16, UI-14 |
| FR-7 | IT-09, API-09 | FR-18 | API-17, API-18, UI-15 |
| FR-8 | IT-12, API-10, UI-09 | FR-19 | UT-SER-01 – 09, API-03, NF-05 |
| FR-9 | IT-10, UT-PRS-25, UT-PRS-26 | FR-20 | UI-01, UI-06, UI-08, UI-11, UI-12, UI-15 |
| FR-10 | IT-11 | FR-21 | UI-02, UI-04, UI-05 |
| FR-11 | NF-04, IT-09 | FR-22 | IT-01, IT-02, IT-04, NF-01, NF-07 |
| NFR-1 | NF-01, IT-01 | NFR-6 | NF-06 |
| NFR-2 | NF-02, UI-02 | NFR-7 | IT-05, IT-06 |
| NFR-3 | NF-03, IT-06 | NFR-8 | NF-08 |
| NFR-4 | NF-04, NF-05, UT-SER-01 – 09 | NFR-9 | IT-05, IT-07, IT-08, IT-09 |
| NFR-5 | API-06, API-07 | NFR-10 | UT-SER-∗, UT-PRS-∗ (no infrastructure required) |

**Table 9** — Requirements traceability matrix.

## 10. Validation Summary

All 38 automated cases pass, with the output reproduced verbatim in Section 11. The 54 documented manual procedures cover every remaining requirement, and each is specified with enough precision — exact commands, preconditions and expected results — that an examiner can re-execute the whole set against a freshly built stack in roughly fifteen minutes. Every functional and non-functional requirement in Chapter 3 of the project report is traced to at least one case in Table 9, with no orphaned requirements and no orphaned tests.

Two defects remain open and are stated plainly rather than smoothed over. D-06 (a genuine zero compute-unit value being recorded as unknown) is a real correctness bug; it is pinned by a test that documents current behaviour so that a future fix is a deliberate, reviewed change. D-09 (an account with no balance rows predating the third migration) is a consequence of forward-only indexing and is documented in the user-facing material. Neither affects the system's core claim, and both are carried forward into the future-work list in the project report, Section 9.2.

# 11. Automated Test Console Output

Captured on 28 August 2026 with `npx vitest run --reporter=verbose` in each package. Reproduced verbatim.

## 11.1 Explorer — `explorer/`

```
 RUN  v4.1.11 /Users/…/Solana-Local-Indexer-Explorer/explorer

 ✓ src/lib/serialization.test.ts > serializeBigInts > returns null/undefined unchanged 1ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > converts a top-level bigint to a string 0ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > converts a Date to an ISO string 1ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > leaves primitives (number, string, boolean)
unchanged 0ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > recursively converts bigints inside a nested
object 1ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > recursively converts bigints inside an array
0ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > handles a realistic API payload shape
(transaction list + pagination) 0ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > handles a negative bigint (balanceChange can
be negative) 0ms
 ✓ src/lib/serialization.test.ts > serializeBigInts > handles zero as a bigint 0ms

 Test Files  1 passed (1)
      Tests  9 passed (9)
   Start at  04:34:26
   Duration  199ms (transform 40ms, setup 0ms, import 54ms, tests 6ms, environment 0ms)
```

## 11.2 Indexer — `indexer/`

```
 RUN  v4.1.11 /Users/…/Solana-Local-Indexer-Explorer/indexer

 ✓ src/grpc/parse-transaction.test.ts > parseSignature > decodes a Buffer signature to base58 2ms
 ✓ src/grpc/parse-transaction.test.ts > parseSignature > decodes a Uint8Array signature to base58
1ms
 ✓ src/grpc/parse-transaction.test.ts > parseSignature > passes through a string signature
unchanged 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseSignature > falls back to 'unknown' for missing
signature 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseSignature > falls back to 'unknown' for an unexpected
type (e.g. number) 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseAccounts > decodes Buffer account keys to base58
strings 1ms
 ✓ src/grpc/parse-transaction.test.ts > parseAccounts > passes through string account keys
unchanged 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseAccounts > returns an empty array when accountKeys is
missing 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseAccounts > preserves account key order (index is used
as accountIndex elsewhere) 1ms
 ✓ src/grpc/parse-transaction.test.ts > parseInstructions > extracts programIdIndex, accounts,
decoded data, and index 1ms
 ✓ src/grpc/parse-transaction.test.ts > parseInstructions > defaults 'accounts' to an empty array
when missing 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseInstructions > sets data to null when instruction has
no data 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseInstructions > assigns instructionIndex matching array
position for multiple instructions 0ms
```

```
 ✓ src/grpc/parse-transaction.test.ts > parseInstructions > returns an empty array when
instructions are missing 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseComputeUnitsUsed > converts computeUnitsConsumed to
BigInt when present 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseComputeUnitsUsed > returns undefined when
computeUnitsConsumed is missing 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseComputeUnitsUsed > returns undefined when
computeUnitsConsumed is 0 (falsy but valid) 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseFee > converts fee to BigInt when present 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseFee > returns undefined when fee is missing 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseSuccess > returns true when meta.err is null 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseSuccess > returns true when meta.err is undefined
(missing) 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseSuccess > returns false when meta.err is a populated
error object 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseError > returns null when there is no error 1ms
 ✓ src/grpc/parse-transaction.test.ts > parseError > serializes an error object to a JSON string
0ms
 ✓ src/grpc/parse-transaction.test.ts > parseBalances > extracts preBalances and postBalances
arrays 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseBalances > defaults to empty arrays when balances are
missing 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseLogs > extracts logMessages when present 0ms
 ✓ src/grpc/parse-transaction.test.ts > parseLogs > defaults to an empty array when logMessages are
missing 0ms
 ✓ src/grpc/parse-transaction.test.ts > End-to-end sample transaction (successful vote tx shape) >
parses every field consistently for a realistic full payload 0ms

 Test Files  1 passed (1)
      Tests  29 passed (29)
   Start at  04:34:31
   Duration  189ms (transform 33ms, setup 0ms, import 47ms, tests 11ms, environment 0ms)
```

**SUMMARY**

**38 of 38 automated tests pass** across 2 test files, in under a second of total execution time, with no validator, database or network dependency.